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Method for Detecting Micron Cracks on a Magnetic Rotor Surface Based on a Support Vector Machine

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ABSTRACT Micro-scale cracks on the surface of a magnetic rotor are difficult to detect because of the formation of stains. A method that is based on computer vision and a support vector machine (SVM) that can accurately and rapidly locate micro-scale cracks are proposed. To further highlight the cracks and reduce interference, the image is pre-processed using improved adaptive median filtering, image sharpening, extract feature point, and an expansion based on mathematical morphology. Next, the feature curves of cracks and stains are extracted as training samples using the feature curve extract algorithm and the region orientation algorithm (ROA) for continuous and discontinuous crack curves images, respectively. SVM is later used to distinguish between the two types of image sources. An improved region orientation algorithm (IROA) is proposed to increase the efficiency of discontinuous crack curves images through online detection. Experiment results show that the accuracy of crack detection using ROA is 97.9%, and the accuracy is 94.7% using IROA. However, the IROA algorithm execution time is 3% that of ROA, and IROA's average crack detection time is shorter than that of ROA by 51%. The proposed method provides the basis for the automatic online detection of micro-scale cracks on a magnetic rotor surface and paves the way for the identification of cracks on complex morphologies, such as split cracks and circular cracks.

INDEX TERMS Micro-scale crack, stain, region orientation algorithm, support vector machines, improved region orientation algorithm, online detection.

I. INTRODUCTION

During the process of magnetization, the surface of an NdFeB magnetic rotor is prone to micron-scale cracks (80 to 150 microns) in the axial direction because of such factors as uneven magnetization. If this cracked magnetic rotor is used in a motor, it can easily cause damage and malfunction, and there is even a risk of explosion. Therefore, a method for detecting the surface cracks on a magnetic rotor has important practical significance in product production.

In the earlier works, many detection methods based on physical characteristics of cracks was proposed, such as used fiber laser ultrasonic sensors to measure acoustic emission generated by cracks in aluminum panels [1], used magnetic eddy current detection (MB-ECT) sensors to detect magnetic conductivity distortion around surface cracks in different magnetized states [2]. These physical methods are performed well for crack detection accuracy on macroscopic cracks, but there will be a serious decline in detection performance on micron cracks. Moreover, device cost to implement these detection methods are expensive, which is not suitable for industrial application.

The automatic detection of micron cracks on magnetic rotor surface is inherently a challenging task. This is because 1) the magnetic rotor on the industrial production line often has stains on the surface due to the production process. These stains will greatly interfere with the detection of micron cracks; 2) the micron cracks are very small, different illumination has a great influence on the image acquisition by the camera, which makes the requirement for the robustness of the crack detection algorithm harsh; 3) cracks on the surface of magnetic rotor are damage of the morphological structure caused by the uneven heating during magnetic charging,

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which leads to irregular morphological topological structure of these cracks.

Recently, many image acquisition systems, such as industrial cameras, scanning electron microscopes, and infrared thermography and computed tomography (CT) scans, have been used as effective monitoring methods for the identification of cracks and defects on surfaces. Cho's research [3] on the influence of illumination and shooting distance on crack image recognition found that as the shooting distance increases, the effects of illumination on crack recognition become greater. It paves the way for micron cracks detection, and help us to find a suitable image acquisition condition for micron cracks image acquisition. On that basis, many crack detection methods based on image processing have been proposed. For example, Talab *et al.* [4] proposed Otsu method and multiple filtering in image processing techniques to detect cracks in cement, this method involves three steps: First; change the image to a gray and using edge of the image and use Sobel's method and development of an image filter using Sobel's filter for detecting cracks. Second; using a suitable threshold in a binary image and classifies all pixels two categorizations background and foreground, and gets the region area after that use filter area and changes the area if less than the specific number to back. Third; after using Sobel's filtering to elimination of residual noise, and detecting major cracks using Otsu method. Experimental work shows that their method is improved relatively to the other methods that aim to detect crack patterns in cement. With the application of complementary-metal-oxide-semiconductor(CMOS) cameras, Zhang *et al.* [5] presented a distance histogram based shape descriptor that effectively describes the spatial shape difference between cracks and other irrelevant objects. Its classification results successfully removed over 90% misidentified objects. Also, compared with the original gray-scale images, over 90% of the crack length is preserved in the last output binary images. As for cracks in the beam structure, a perturbation/slop singularity is induced in the deflection profile, a high sensitive crack detection was proposed on a beam under certain deflection by optimizing spatial wavelet analysis [6]. Spatial wavelet transformation works as a magnifier to amplify the small perturbation signal at the crack location to detect and localize the damage. The profile of a deflected aluminum cantilever beam is obtained for both intact and cracked beams by a high resolution laser profile sensor. Gabor wavelet transformation is applied on the subtraction of intact and cracked data sets. Their method can detect a small crack with depth of less than 10% of the beam height.

Beamlet transform has been widely used for extracting crack features from images, which is an excellent multi-scale geometric analysis method. However, it has a major drawback that it always performs too slowly due to very much redundant computation. In order to solve this problem, Lin and Zhao [7] presented a fast beamlet transform for crack detection. Image semantic information is taken into account in the process. IF is defined for recursive partitioned boxes and vertices on sides of every box. The recursively partitioning process is directed by IF. Furthermore, geometric structures, which have been used in grouplet transform, are introduced. Geometric flows of an image are determined by computing association field values at every point. Beamlets in every box are reduced according to the major direction of geometric flows.

Those methods mentioned above based on image processing performed well in detection accuracy, however, they required relatively more time to complete detection which makes them unfit for industrial applications. In order to improve the speed of crack detection, Qu *et al.* [8] proposed an improved algorithm for image crack detection, which included an accelerated algorithm and a new denoising method based on the percolation model. The accelerated algorithm decreases the number of iterations of percolation processing to reduce the computing time, and the denoising method is based on the characteristic of brightness and the length feature of cracks to remove the noisy regions. Also, Yamaguchi and Hashimoto [9] introduced an efficient and high-speed crack detection method that employs percolation-based image processing. They proposed termination and skip-added procedures to reduce the computation time. The percolation process is terminated by calculating the circularity during the processing. Moreover, percolation processing can be skipped in subsequent pixels according to the circularity of neighboring pixels. Although these methods proposed by them has improved the speed of the crack detection algorithm, it is not enough for real industrial inspection application.

With the development of the computing capabilities and image acquisition equipment, many machine learning algorithms such as convolutional neural network (CNN) [10]–[12], support vector machine (SVM) [13], [14], Random Forests (RF) [15], have been introduced into pavement crack detection and achieved acceptable results. However, most of these algorithms detect cracks in block-level, i.e. they first partition images into pixel blocks and detect a block as crack block if it contains crack pixels. However, above approaches neglect the actual borders and the widths of the cracks to be detected, thus do not accurately measure cracks for subsequent applications, and also can't detect the path of cracks.

In this paper, an automatic method to rapidly detect micro-scale cracks on the surface of a magnetic rotor was proposed. First, the acquired image was extracted as a feature point map through image preprocessing to simplify the calculation. Next, FCEA and ROA were used to obtain a complete feature curve that represents the spatial location and morphological features of cracks and stains. Simultaneously, the length and distribution of the crack feature curves were counted, and the speed of ROA was improved based on the number and position of the optimal representative arcs that were obtained from the statistical data. After that step, the crack feature curve was used as a positive example, and the stain feature curve was used as a negative example to train the SVM.

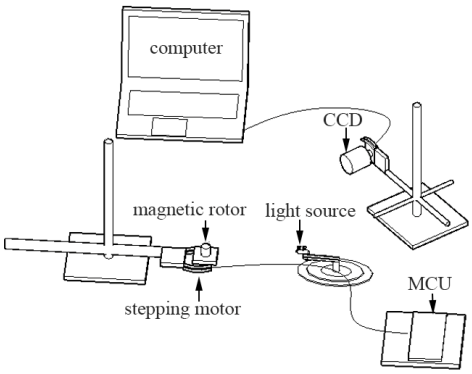


FIGURE 1. Concept scheme of image acquisition system.

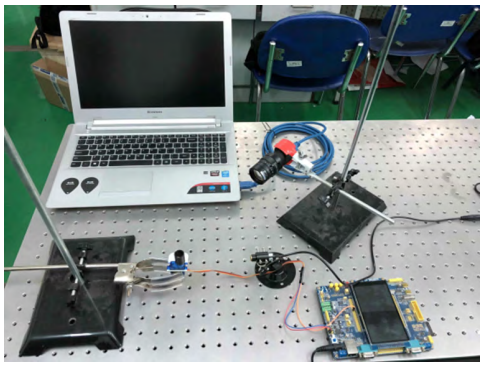


FIGURE 2. Physical picture of image acquisition system.

Finally, the trained SVM classification was used to classify the feature curve. Our proposed method also has potential to detect macroscopic cracks, such as cracks in steel and bearings. In future work, we will conduct further research on this aspect.

II. IMAGE ACQUISITION SYSTEM

A. SYSTEM DEVICE DIAGRAM

Image acquisition system is shown in Fig. 1 and Fig.2:

The computer was equipped with a CORE i7 processor (2.4 GHz) and 8G memory; the CCD model is MER-231-41GM/C-P with resolution ratio 1920×1200 and was produced by the Daheng Invision Group. The MCU adopted stm32F4ZGT6, and yellow highlight LED was used as a light source. The MCU was used to control a stepper motor to rotate the magnetic rotor for image acquisition (each 60° one image).

B. MAGNETIC ROTOR SAMPLES

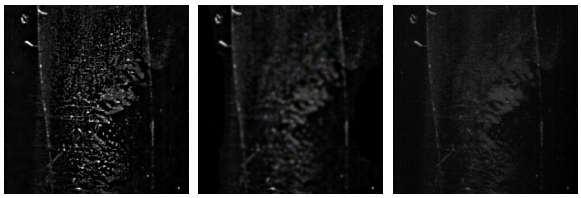
Three hundred images of magnetic rotors with cracks were selected for the experiment. Because the cracks on the magnetic rotor surface were very fine, a special auxiliary light source was required in order to see the crack with the naked eye. Selected sample images are shown in Fig.3, and Fig. 3 (a) is the image of a normal magnetic rotor that is illuminated by the special light source. The morphological characteristics of a normal magnetic rotor are shown in Table 1.

(a) (b) (c)

FIGURE 3. Image of a magnetic rotor; (a) Image of a normal magnetic rotor, (b) and (c) Image of two magnetic rotors with cracks.

TABLE 1. Morphological characteristics of a magnetic rotor.

Diameter (mm)	Thickness (mm)	Height (mm)	Volume (cm^3)
16.0	2.0	22.5	18.45



(a) (b) (c)

FIGURE 4. Comparison of the processing effects of the traditional adaptive median filtering algorithm and the IAMF algorithm: (a) Grayscale of the original image; (b) Processing effect of the traditional adaptive median filtering algorithm; (c) Processing effect of IAMF algorithm.

III. IMAGE PRE-PROCESS

Due to defects in the manufacturing process, there will always be stains on the surface of the magnetic rotor that will interfere with the extraction of the crack feature curves. The image was pre-processed using an improved adaptive median filter (IAMF) to reduce image interference. A Harris Corner Detector was used to extract image feature points.

The datasets of the image are denoted as the following:

$$P_{img} = \{(x_i, y_j, z_{ij})\}, \quad 1 \le x_i \le 1920, 1 \le y_j \le 1200, \quad 0 \le z_{ij} \le 225 \tag{1}$$

where x and y denote the pixel coordinates and z represents the gray value of the pixels.

A. IMPROVED ADAPTIVE MEDIAN FILTERING (IAMF)

The magnetic rotor surface with cracks and stains is shown in Fig.4. The average gray value of a crack curve is different from the stain in a sample area. The crack curve is slender, and its corresponding average gray value is small. The stain is dense, and its corresponding average gray value is larger than that of the crack curve. Fig.8 (a) shows the 3×3 sample matrix that was used to sample P_{img}. If the average gray value of the sample matrix was greater than a particular threshold, then this indicates that the pixel points of the sampling matrix are stains and following steps were used. Otherwise, not used them.

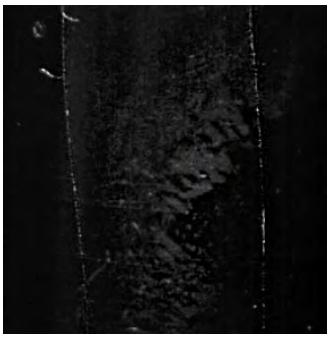


FIGURE 5. Processing effect of image sharpening from Fig. 4 (c).

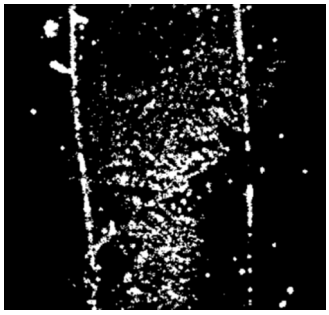


FIGURE 6. Effect of feature point extraction from Fig. 5.

Step1: Use the 3×3 x-shaped window as IAMF initial operator and calculated median gray value Z_{med}, the maximum gray value Z_{max}, and the minimum gray value Z_{min} of pixels in this window.

Step2: If $Z_{med} = Z_{min}$ or $Z_{med} = Z_{max}$ then automatically increases the size of the x-shaped window and repeat Step2.

Step3: If the size of the x-shaped window reach 9×9 then the original pixel will not be modified and skip Step4.

Step4: Use the original pixel value to judge Z_{max} and Z_{min}. If the pixel value is equal to Z_{max} or Z_{min}, then use Z_{med} instead of its original value.

The difference between the traditional adaptive median filtering algorithm and IAMF is shown in Fig. 4. From Fig. 4, we can see that IAMF and the conventional adaptive median filtering algorithm can effectively suppress the stains, but IAMF has a better effect on crack fidelity.

B. IMAGE SHARPENING

To highlight the crack on the surface of the magnetic rotor, the USM algorithm was used. The effect is shown in Fig. 5.

C. EXTRACT FEATURE POINT

To prepare for the feature curve extraction, a Harris Corner Detector was used to extract the feature points of the image to eliminate the information in P_{img} that has little coupling with the crack. The extracted feature points are shown in Fig. 6. As seen from Fig. 6, most of the feature points of the cracks have been extracted. However, after the corner detection, the image has many scattered feature points because of stains.

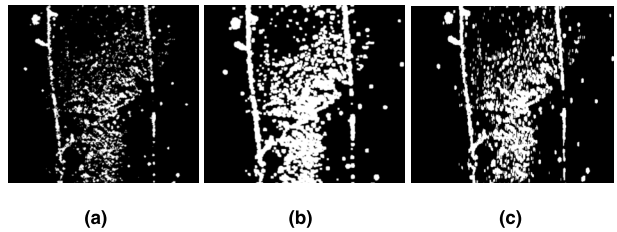


FIGURE 7. Comparison of the processing effects of traditional expansion algorithms and expansion based on mathematical morphology; (a) Image of feature points, (b) Processing effect of traditional expansion algorithm, (c) Processing effect of expansion based on mathematical morphology.

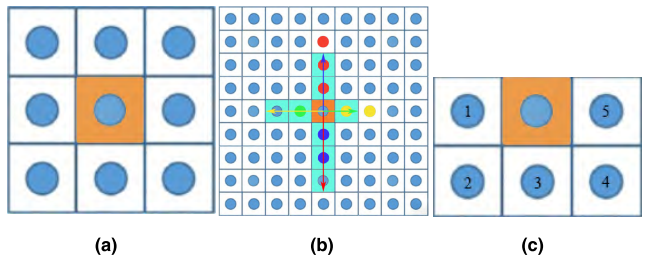


FIGURE 8. Operators of IAMF, expansion algorithm, and FCEA: (a) Operator of IAMF; (b) operator of the expansion algorithm; (c) operator of the FCEA that are used for finding the next target feature point.

D. EXPANSION BASED ON MATHEMATICAL MORPHOLOGY

In the corner detection process, the gradient of the gray value of certain of the pixels on the crack did not reach the threshold value and were filtered by the computer; thus, they did not appear as feature points. This situation manifested as a discontinuity of the crack (a fracture of the crack curve), which affected the integrity of the crack feature curve extraction. Therefore, the expansion algorithm that is based on the mathematical morphology was used to patch a part of the filtered pixels. The algorithm's working principle is expressed as follows:

Select a of 9×9 pixel matrix around a certain feature point, as shows in Fig. 8 (b). Calculate the number of feature points in the direction of the arrows, as indicated by the four different colors. The expansion distance is determined by the number of feature points of the opposite direction, the shape of the expansion operator algorithm is obtained. The cyan region in Fig. 8 (b) represents the operator applied to the dilation operation. Expansion based on the mathematical morphology processing effect is shown in Fig. 7 (c). The continuity of the cracks is better in Fig. 7 (c), although the size of the stain area has also increased. However, the features of stains and cracks are not the same and will be distinguished in the subsequent classification steps.

IV. COMPLETE FEATURE CURVE EXTRACTION

A. FEATURE CURVE EXTRACTION ALGORITHM (FCEA)

The steps of this algorithm are as follows:

Step1: As shown in Fig. 8 (c), the middle orange point is set as the target feature point, and the numbers 1 to 5,

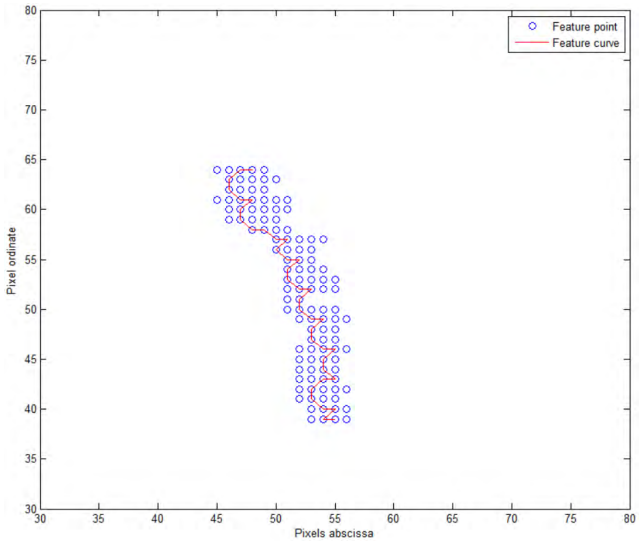
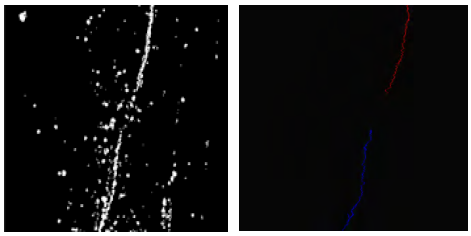


FIGURE 9. Extraction process effect of FCEA.



(a) (b)

FIGURE 10. Effect diagram of FCEA: (a) feature point map; (b) processing effect of the FCEA, at which different colors indicate different cracks that are identified by the FCEA.

representing the 5 pixel points for which the algorithm looks for in sequence. If a feature point exists, it is added to the feature curve. Next, the feature point is updated to the new target feature point.

Step2: Step 1 is repeated to find the next target feature point until no feature points that meet the conditions are found.

Step3: Step1 and Step2 are repeated until all the feature points in P_{img} are recognized by the FCEA.

Since the length of the crack feature curve that is extracted from cracks by the FCEA is longer than the length of most of the interference feature curves that are extracted from stains, a feature curve length threshold is set to filter out most of the interference curves extracted from the stains. The threshold is set to 40 pixels and is denoted by l_{ths}.

Fig. 9 and Fig. 10 show the effect of the FCEA extraction feature curve. The extraction of the feature curve greatly simplifies the information in P_{img}, since the coupling with the crack is weak.

B. REGION ORIENTATION ALGORITHM (ROA)

As shown in Fig. 13 (a), further observation of the feature curve extracted by the FCEA revealed that it was partially distorted with respect to the feature point map in Fig. 13 (c).

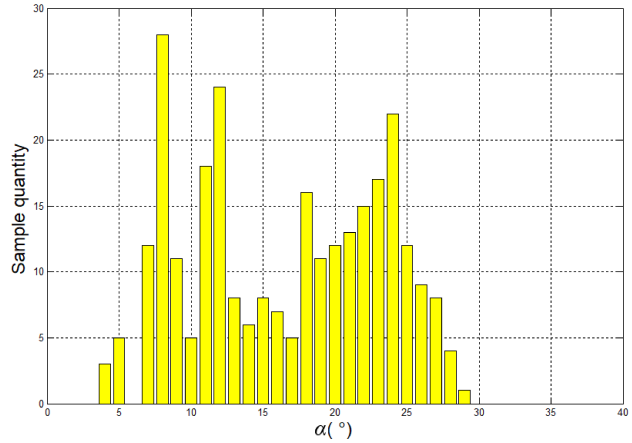
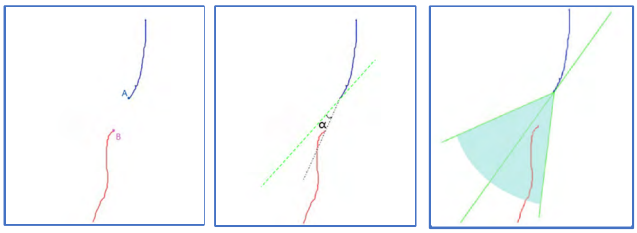
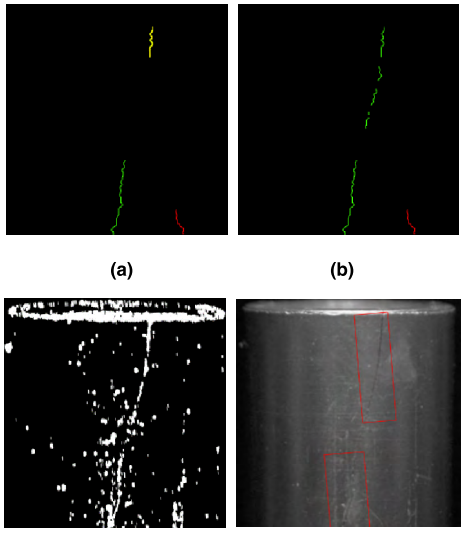


FIGURE 11. Values distribution of α.



(a) (b) (c)

FIGURE 12. (a) Schematic diagram of the end-point of the feature line. (b) Schematic diagram of a definition. (c) Schematic diagram of ROA search area.



(a) (b) (c) (d)

FIGURE 13. Comparison of the effects of FCEA and ROA: (a) Effect of only used FCEA; (b) effect of ROA combined with FCEA; (c) feature point map; (d) original image.

This finding is observed because a short feature curve had been filtered out without reaching l_{ths}; however, if l_{ths} had been smaller, there would have b no reason to reject the interference feature.

When a crack occurred in the feature curve upon an extraction by FCEA, a method to reconnect the feature curve was proposed: the region orientation algorithm (ROA). ROA explores a certain fan-shaped region along the tangent direction at the end of the feature curve when FCEA completes the extraction. The algorithm detects whether there is a feature curve that can be connected.

The curve using the ROA is denoted by *line*, and *line* can be represented as a set of feature points:

$$line = \{(x_1, y_1, z_1), (x_2, y_2, z_2), (x_{l-1}, y_{l-1}, z_{l-1}), (x_l, y_l, z_l)\} \quad z = 255 \tag{2}$$

where x is the abscissa of the pixel, y is the ordinate of the pixel, z represents the gray value of the pixel.

The formula for calculating the angle of the tangent at the end of the feature curve is defined as the following:

$$\beta = \sum_{n=1}^{k} \frac{1}{k} \cdot \arcsin \frac{y_{l-n+1} - y_{l-n}}{\sqrt{(x_{l-n+1} - x_{l-n})^2 + (y_{l-n+1} - y_{l-n})^2}} \quad (-180^\circ \le \beta \le 0^\circ) \tag{3}$$

where β is the angle of the tangent at the end of the feature curve and k is the number of pixels at the end of the feature curve used to calculate the tangent line. In this experiment, k equals 10.

The coordinates of the end-point (x_e, y_e) of the *line*, as shown at point A in Fig. 12 (a), and the coordinates of the starting point of the crack to be connected are denoted by (x_s, y_s) and are denoted as point B in Fig. 12 (a). As shown in Fig. 12 (b), α is the angle between the two-point line segment between A and B and the tangent line at the end of *line*. The definition of α is as follows:

$$\alpha = \left| \arcsin \frac{y_e - y_s}{\sqrt{(x_s - x_e)^2 + (y_s - y_e)^2}} - \beta \right| \quad (0^\circ \le \alpha \le 180^\circ) \tag{4}$$

Next, use FCEA to collect several crack feature curves and count 300 α values at random to see α distributions. The α distributions are shown in Fig. 11.

In Fig. 11, the numerical values of α are distributed from 0° to 30°. Therefore, the fan-shaped region searched by the ROA is set as a fan shape with an opening angle of 60°. Therefore, the tangent at the end of the *line* is symmetrical, as shown in the light cyan region in Fig. 12 (c). The crack detection effect using ROA is shown in Fig. 13 (b). The algorithm of ROA is shown in algorithm 1. The default sector radius of the ROA is 120 pixels.

V. FEATURE CURVE CLASSIFICATION

After the feature curve extraction was completed, as shown in Fig. 14, we needed to distinguish between the crack feature curve and the interference feature curve (from stain). To solve this problem, SVM was used as a classifier, and eight characteristics of the feature curve were used as the classification basis. Next, 300 crack curves and 300 interference curves were selected.

Algorithm 1 Algorithm for ROA

Input: P_{img} processed by FCEA;
Output: feature curve image P_d;
1. **begin**
2. **foreach** $line \in P_{img}$ **do**
3. **T**: $line_{temp} \leftarrow line$;
4. **for** $t \leftarrow 1$ **to** 10
5. $\beta \leftarrow \beta + \arcsin\left((y_{l-t+1} - y_{l-t}) / \sqrt{(x_{l-t+1} - x_{l-t})^2 + (y_{l-t+1} - y_{l-t})^2}\right)$;
6. **end for**
7. $\beta \leftarrow \beta/10$;
8. $R_{area} \leftarrow$ *all pixels of the 60° sector region determined by* β;
9. **foreach** *pixel* v *in* R_{area} **do**
10. **if** $v \in line_{exi}$ /* $line_{exi}$ is another *line* in P_{img}*/
11. $line \leftarrow line \cup line_{exi}$;
12. **end if**
13. **end foreach**
14. **if** $line \neq line_{temp}$
15. **goto** T;
16. **end if**
17. **end foreach**
18. $p_d \leftarrow p_{img}$;
19. **end**

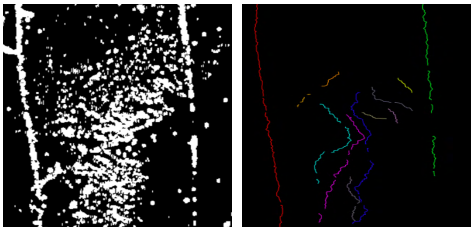


FIGURE 14. Effect of feature curve extraction: (a) Feature point map; (b) extracted feature curve, where different colors indicates different feature curves.

A. SUPPORT VECTOR MACHINE

The fundamental idea of the SVM is to construct a hyperplane as the decision line that separates the classes with the largest margin (introduced by Cortes and Vapnik [16] (1995)).

The SVM is based on the theory proposed by Pal and Foody [17] (2010) that feature selection can improves classification accuracy . In this paper, there are v ($v = 300$) crack feature curve samples in the training data corresponding to two groups. Each sample is denoted by a vector S_i ($i = 1, \ldots, v$), and each vector has 8 items that represent the selected crack features. In addition, a vector $y \in (-1, 1)$ denotes the two classes of stain and crack, respectively. The input vector S_i is mapped into a high-dimensional feature space by using a suitable kernel function $k(u_i, u_j)$ for non-linear training data. The two kernels used in this study, the Gaussian radial basis function (rbf) and the linear

function, are defined mathematically by

$$\begin{cases} k_{rbf}(X_i, X_j) = \exp(-\dfrac{||X_i - X_j||^2}{2\sigma^2}) \\ k_{liner}(X_i, X_j) = X_i^T X_j \end{cases} \tag{5}$$

where k_{rbf} is the Gaussian radial basis function kernel, and k_{liner} is the linear kernel.

The classification function, $f(x)$, is defined using training data and will be used to classify the unseen test data set. This function is defined in terms of kernels:

$$f(x) = \mathrm{sgn}[\sum \alpha_i q_i k(X_i, X_j) + b] \tag{6}$$

where b is a bias term and α_i is the Lagrange multiplier coefficient obtained by solving the quadratic programming problem (introduced by Burges *et al.* [18] (1998)).

$$Maximiz : W(\alpha) = \sum_{i=1}^{v} \alpha_i - \frac{1}{2} \sum_{i=1}^{v} \sum_{j=1}^{v} q_i q_j k(X_i, X_j) \alpha_i \alpha_j \tag{7}$$

This equation must satisfy the following constraints: $0 \leq \alpha_i \leq C$ $(i = 1, \ldots, v)$ and $\sum_{i=1}^{v} q_i \alpha_i = 0$, where C is a nonnegative regularization parameter.

B. FEATURES EXTRACT

Employing 8 parameters as features to judge, based on shape, whether a feature curve is a crack is complicated. It is difficult to fully represent a crack using a single feature. However, by utilizing more features, one will not obtain a better result because certain features are redundant, and others, such as a feature's weight, vary.

We note that all crack feature curves in the image must be identified and numbered. The numbering rule is from left to right and from top to bottom. The midpoint of the crack feature curve is denoted by M, and the corresponding image coordinates are (M_{xi}, M_{yi}). The pixel on the crack feature curve is denoted by p, and p_i denotes the i-th pixel on the feature curve.

$Crack_D$: The total number of feature curves present in a circle with radius R centered on the midpoint M of the crack N_c divided by the area with R set equal to l_{ths}.

$$Crack_D = \frac{N_c}{\pi R^2} \tag{8}$$

$Crack_L$: The sum of the number of pixels in the feature curve.

Inclination: The value of α.

Area: The number of feature curves in the circle with the distance between the middle points of the two number adjacent feature curves as the radius and the midpoint of the line connecting the midpoints of the two feature curves as center.

$Maximum_d$: The maximum distance between pixels in the feature curve.

$Dist_m$: The average distance of all points on the feature curve to the midpoint.

$$Dist_m = \frac{1}{Crack_L} \sum ||p_i - M|| \tag{9}$$

$Dist_d$: The deviation of the distance from the pixel of the feature curve to the center.

$$Dist_d = \sqrt{\frac{1}{Crack_L} \sum (||p_i - M|| - Dist_m)^2} \tag{10}$$

Roundness: The relationship between mean value and standard deviation.

$$Roundness = 1 - \frac{Dist_m}{Dist_d} \tag{11}$$

The extracted crack feature is stored in a two-dimensional array in which each row represents the sequence number of each feature curve and each column represents the eight features of each feature curve.

VI. IMPROVED REGION ORIGIONAL ORIENTATION ALGORITHM (IROA)

A. MAIN FACTORS THAT AFFECT RUNNING EFFICIENCY

To determine the significant factors that affect the speed of the detection, we must calculate the time spent by FCEA and ROA in the application process. We make the following definition: The time for processing the unit pixel of FCEA is denoted T_c; ROA processing unit pixel time is denoted T_o; the sum of the pixel points of the FCEA called for processing an image is denoted I_{con}; and the sum of the pixels of the ROA called for processing an image defined as I_{ori}.

The remaining processing time, e.g., image acquisition, reading, preprocessing, are denoted T_{fix}. The time T for processing a picture defined as

$$T = T_c \cdot I_{con} + T_o \cdot I_{ori} + T_{fix} \tag{12}$$

The data from the statistical analysis of the FCEA and ROA calling time in the image processing process is shown in Fig. 15. We can see from Fig. 15 that the computation speed for crack detection using ROA is relatively slow; thus some targeted optimizations of ROA calculations will be required to speed up crack detection.

B. PRINCIPLE OF IROA

To solve the problem of large ROA calculations, the processing of the fan-shaped areas is turned into multiple representative arcs. Because a feature curve has a certain length, it intersects with the representative arc in the fan-shaped region.

However, this calculation method also has problems. As shown by the feature curve indicated by the arrow in Fig. 16, certain of the short-length feature curves may be undetectable with this method. This would result in a decrease in the integrity of the feature curve extraction process. To solve this problem, we need to find an optimal combination solution that is based on the actual number and position of representative arcs. To determine this optimal solution, 300 processed sample images were obtained and counted using a distribution rule, as shown in Fig. 17. The relationship between the arc radius and the number of samples on which the cracks are detected approximates a Gaussian distribution,

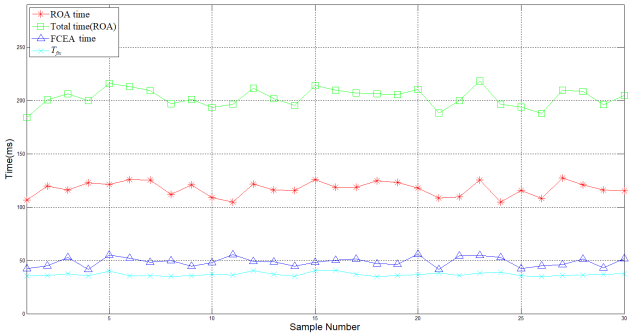


FIGURE 15. Time required for each part of the crack detection process.

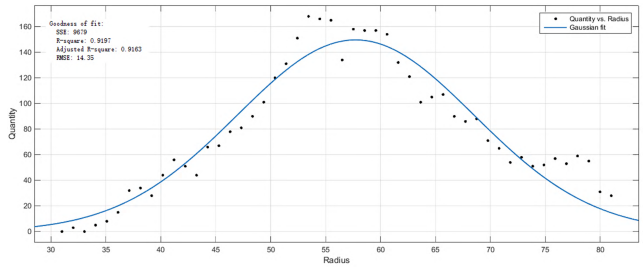


FIGURE 17. Statistical distribution of samples.

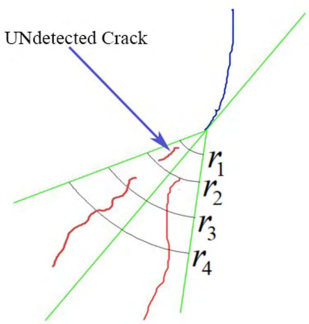


FIGURE 16. IROA schematic diagram.

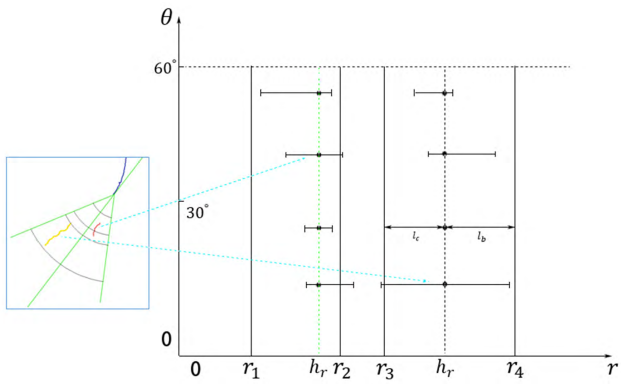


FIGURE 18. Schematic diagram of the detection principle of IROA. h_r is an arc radius with an undetermined value.

also shown in Fig. 17. Therefore, in the following theoretical analysis, we suppose that the relationship between the arc radius and the probability of detecting the feature curve on the arc obeys the Gaussian distribution.

The probability that the feature curve can be detected on this arc is the following:

$$f(h_r) = \frac{\frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(h_r-\mu_1)^2}{2\sigma_1^2}}}{\int_0^\infty \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(h_r-\mu_1)^2}{2\sigma_1^2}}} \tag{13}$$

where h_r is the radius of the representative arc.

Suppose that the ROA maximum exploration radius, denoted by r_{max}, is divided into n representative arcs. Next, the total number of pixels calculated by ROA satisfies the following formula:

$$\begin{cases} I_{ori} = \sum_{i=1}^{r_{max}} \frac{\pi}{3} \cdot i \\ I_{ior} = \sum_{i=1}^{n} \frac{\pi}{3} \cdot r_i \end{cases} \tag{14}$$

where I_{ori} is the sum of the pixels of an image processed by ROA, I_{ior} is the sum of the pixels of an image processed by IROA, and r_i is the arc radius of each representative arc.

By equation (13), we are now able to compute the probability that the feature curve can be detected on the representative arc of a certain radius. If we can compute the length of the crack that appears on this arc radius, then the theoretical optimal combination solution corresponding to the maximum crack detection rate can be computed by using a limited number of representative arcs. Since the length of the crack is random, the statistical analysis method can be applied to the lengths of the feature curves of the 300 samples. The final analysis results are similar to the position distribution rule of the feature curve, and the length of the feature curve is also consistent with the Gaussian distribution. We assume that the length of the feature curve and the radius of the representing arc are independent of each other. Therefore, the probabilistic formula for deriving the feature curve on the arc is defined as:

$$g(h_l) = \frac{\frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(h_l-\mu_2)^2}{2\sigma_2^2}}}{\int_0^\infty \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(h_l-\mu_2)^2}{2\sigma_2^2}}} \tag{15}$$

where h_l is length of the feature curve on the arc.

Fig. 18 shows two feature curves and their detection principle under four representative arc distributions. If there is a feature curve on the arc h_r, then the distance from the closest representative arc to arc h_r is the detectable minimum critical distance l_c, and the length of the crack must be greater than l_c to have an intersection with the representative arc to be detected. The second nearest representative arc from h_r is denoted by l_b. l_c, l_b are calculated as follows (Note: The following derivation is performed in the coordinate system shown in Fig. 18):

$$l_c = \min\{|h_r - r_i|\} \quad (1 \le i \le n) \tag{16}$$

$$l_b = \min\{|h_r - r_j|\} \quad (1 \le j \le n \text{ and } j \ne i) \tag{17}$$

TABLE 2. Theoretical correctness and calculation time when n = 3.

r_{max}	r_1	r_2	r_3	Det	T_{IROA}(ms)	T_{ROA}(ms)
50	16	28	50	69.75%	1.35	18.30
100	21	51	100	83.98%	2.47	72.44
180	39	87	180	67.35%	4.38	233.26

TABLE 3. Theoretical correctness and calculation time when n = 5.

r_{max}	r_1	r_2	r_3	r_4	r_5	Det	T_{IROA}(ms)	T_{ROA}(ms)
60	9	15	25	32	60	84.58%	2.02	26.23
120	25	47	71	94	120	94.20%	5.26	104.12
200	34	57	101	144	200	87.39%	7.69	288.23

TABLE 4. Theoretical correctness and calculation time when n = 8.

r_{max}	r_1	r_2	r_3	r_4	r_5	r_6	r_7	r_8
80	5	16	27	36	45	51	67	80
150	7	19	31	51	75	99	112	150
240	15	35	58	97	134	171	206	240

Det	T_{IROA}(ms)	T_{ROA}(ms)
97.08%	4.69	46.51
97.53%	7.80	162.41
93.35%	13.71	414.78

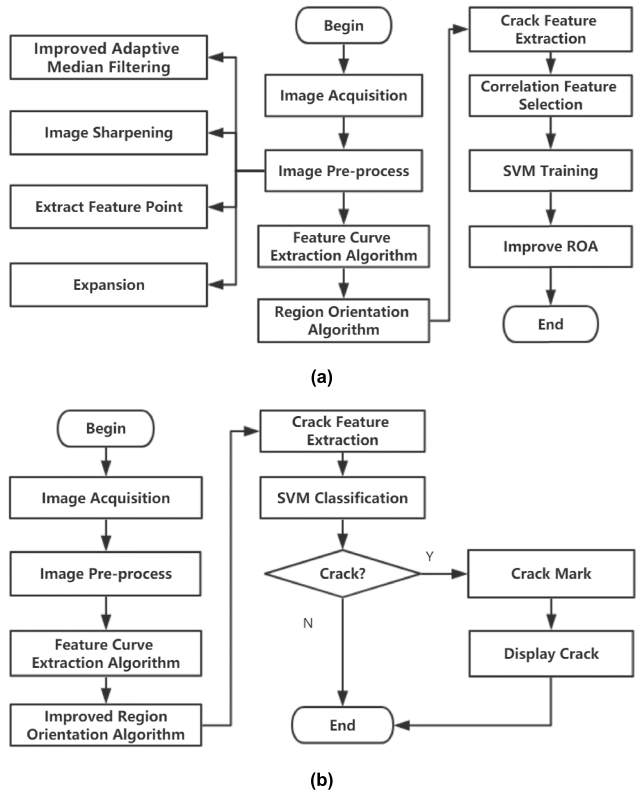


FIGURE 19. Experiment flowchart: (a) Flowchart of samples extraction and SVM training; (b) Flowchart of crack detection using completed SVM.

The probability that a feature curve that appears on arc h_r is not detected is:

$$H(h_r) = \int_0^{l_c} g(h_l)dh_l + \int_{l_c}^{l_b} \frac{l_c}{h_l} \cdot g(h_l)dh_l + \int_{l_b}^{l_c+l_b} \frac{l_c + l_b - h_l}{h_l} \cdot g(h_l)dh_l \quad (18)$$

When the search range of the ROA is turned into n representative arcs, the probability that the feature curve in the search range of the IROA is not detected is: $\int_0^{r_{max}} H(h_r) \cdot f(h_r)dh_r$, the detection rate of the feature curve Det is:

$$Det = 1 - \int_0^{r_{max}} H(h_r) \cdot f(h_r)dh_r \quad (19)$$

In this experiment, $u_1 = 56.20$, $\sigma_1 = 21.96$, $u_2 = 32.61$, $\sigma_2 = 56.20$. Table 2, Table 3, and Table 4 show the theoretical correctness and theoretical calculation time for different combinations of representative arcs: where T_{IROA} is the theoretical running time of IROA of each image, T_{ROA} is the theoretical running time of ROA of each image.

In above tables, we find that the running time of the IROA algorithm is considerably shorter than that of the ROA algorithm. We also find that different parameters of the representative arcs cause the running time T_{IROA} and the Det of the IROA algorithm to change. Therefore, we need to find a set of optimal representative arc parameters to balance Det and T. For details, see section C of chapter VII.

VII. RESULTS AND DISCUSSION

A. EXPERIMENT FLOWCHART

The experiment flowchart is shown in Fig. 19:

B. CRACK DETECTION ACCURACY

After the processing steps, certain of the images were converted into feature curves as training samples. In addition, 300 crack curves and 300 interference curves were manually selected as input into SVM for training (the crack curve has a value of '1'; the interference curve has a value of ' −1'). After training is completed, the SVM can be used to classify the feature curves. Table 5 shows the accuracy of the classification of two different kernels and their average crack detection time T_{ave}. This table also shows the effect of the expansion on the detection accuracy. To further analyze the results of the experiment, the confusion matrix was calculated (shown in Table 7) of the SVM classifier result. In Table 6, each column of the confusion matrix represents a prediction of a class, and each row represents an instance of an actual class. In artificial intelligence, the confusion matrix is a visualization tool that is used for supervised learning.

As shown in Table 7, when used 'rbf' kernel, two stain samples are classified as crack, and three crack samples

TABLE 5. Comparison of crack detection with linear kernel and rbf kernel.

Input	SVM Kernel (rbf)	SVM Kernel(linear)	T_{ave}(ms)
P_{img}	97.9%	98.6%	206
P_{img} without expansion	91.7%	89.8%	194

TABLE 6. Four possible scenarios for classification.

Actual	Predicted	
	Crack(1)	Stain (-1)
Crack(1)	True Crack_ Crack(TCC)	False Crack_ Stain(FCS)
Stain(-1)	False Stain_ Crack(FSC)	True Stain_ Stain(TSS)

TABLE 7. Confusion matrix obtained with the linear kernel and the RBF kernel.

Input	Confusion Matrix
P_{img} with 'rbf'	$\begin{pmatrix} 68 & 3 \\ 2 & 56 \end{pmatrix}$
P_{img} with 'linear'	$\begin{pmatrix} 102 & 7 \\ 1 & 68 \end{pmatrix}$

TABLE 8. P and R of using linear kernel and rbf kernel.

Mode	P	R
P_{img} with 'rbf'	97.1%	95.8%
P_{img} with 'linear'	99.0%	93.6%

are classified as stain. When used 'linear' kernel, only one stain sample is classified as crack, but seven crack samples are classified as stain. To compare the results of the two classifier models more clearly, the results of the confusion matrix are used to calculate the precision and recall of the models.

Positive predictive value (Precision):

$$P = \frac{TCC}{TCC + FSC} \tag{20}$$

True positive rate (Recall):

$$R = \frac{TCC}{TCC + FCS} \tag{21}$$

In general, the precision and recall rates are inversely related variables. When the precision rate is high, the recall rate is low. The results are shown in Table 8. In the process of detecting cracks on a magnetic rotor, it is more desirable to minimize the misrepresentation of cracks. Therefore, the recall rate is more important than the precision rate, and the 'rbf' kernel is chosen.

C. ANALYSIS OF CRACK FEATURE PARAMETER

To analyze the correlation of the different features and evaluate which feature has the greatest influence on the result,

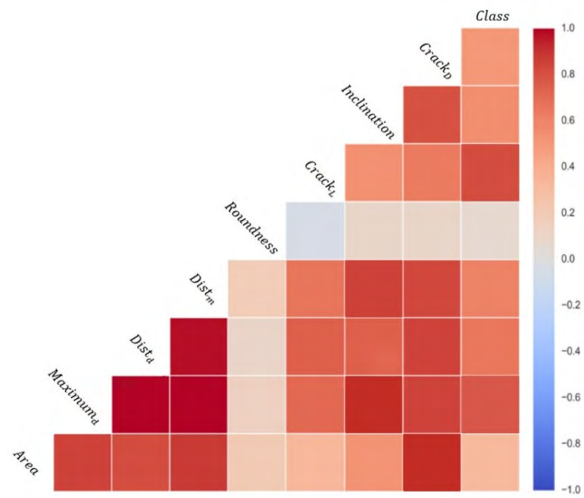


FIGURE 20. Relationship among 8 features and the effect of each feature on the classification results. The grid gets deeper red as the correlation increases.

TABLE 9. Optimal parameter of representative arc.

r_{max}	r_1	r_2	r_3	r_4	r_5	r_6	Det	T(ms)
178	20	39	71	90	120	147	95.0%	6.98

we use the Correlation Feature Selection (CFS) method and plot the correlation matrix as shown in Fig 19. The CFS measure evaluates subsets of features based on the following hypothesis: Desirable feature subsets contain features that are highly correlated with the classification and that are uncorrelated to each other.

As we can see in Fig. 20, the far right column represents the relationship between the different features and the class (crack or stain). As the correlation increases, the color of the grid becomes deeper red. We observe that the feature *Roundness* has minimal effect on the results and that the feature $Crack_L$ is strongly correlated with *Class*. Furthermore, these three features ($Maximum_d$ and $Dist_d$, $Dist_m$) are strongly correlated.

D. ALGORITHM SPEED OPTIMIZATION ANALYSIS

IROA improves the algorithm speed by sacrificing the accuracy of crack detection. Thus, a balance between *Det* and *T* needs to be found. To solve this problem, the simulated annealing algorithm proposed by Turanoğlu and Akkaya [19] (2018) was used; however, we require that the *Det* reach 95% to obtain the optimal representative arc combination solution, as shown in Table 9. Fig. 20 illustrates the running time of the algorithm using IROA and ROA.

With Fig. 21 and Table 9, we observe that the average running time of IROA is approximately 3% of ROA, and the average crack detection time of IROA is 51% shorter than that of ROA. Therefore, the goal of reducing the running time of the algorithm is achieved.

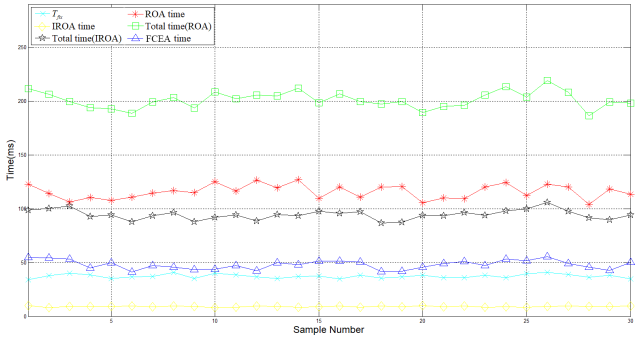


FIGURE 21. Time required for each part of the crack detection process.

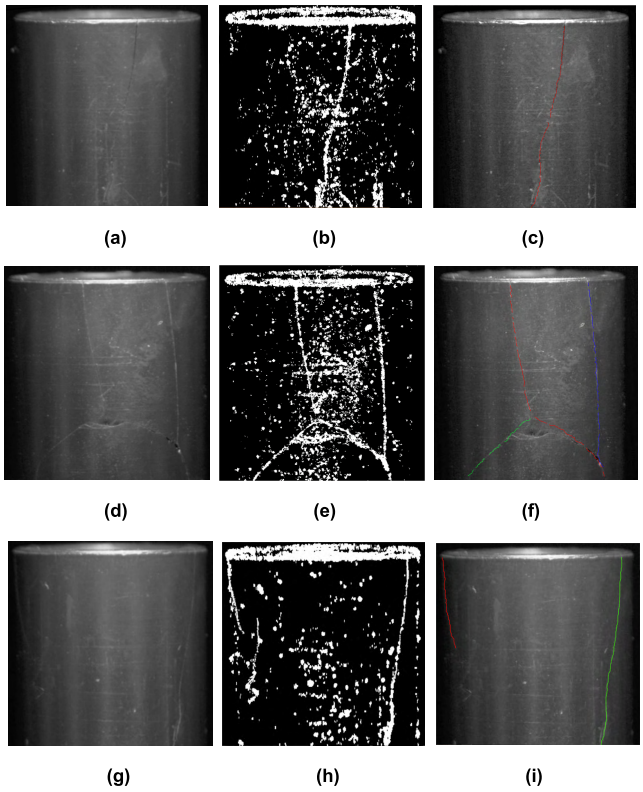


FIGURE 22. Effect of different cracks detection; (a), (d), (g) Original images, (b), (e), (h) feature point maps after expansion, (c), (f), (i) display detected cracks.

E. SVM CLASSIFIER TEST RESULTS

Different colors are used in Fig. 22 to distinguish different cracks and to superimpose the identified cracks on the original image. After testing 300 crack magnetic rotors, we determined that the accuracy of crack detection is 97.9% with ROA, and the detection accuracy rate of fracture cracks is 94.7% with IROA. As shown in Fig. 22 (a)(b)(c), (d)(e)(f), (g)(h)(i), the fracture cracks in both the original image and the feature point map are determined to be complete cracks, which overcomes the original problem with the conventional crack identification method. Fig. 22 (f) shows that our crack detection system can distinguish among cracks with different paths, a result that provides a good basis for further study of the path of the crack curve.

TABLE 10. Comparison of crack detection method in running time and accuracy.

Method	ROA	IROA	Qu	Yamaguchi	CNN
Accuracy(%)	97.9	94.7	81.3	86.0	97.4
T(s)	0.197	0.097	0.122	1.01	2.55

Comparison of crack detection accuracy and time among our proposed methods and other methods with similar functions is given in Table 10. It proves that our methods have better performs than other methods in detection accuracy and speed. where ROA and IROA are our proposed methods, Qu *et al.* [8] and Yamaguchi and Hashimoto [9] and CNN [12] are methods proposed in other papers.

VIII. CONCLUSIONS

This paper proposed several methods to rapidly and accurately identify micro-scale cracks on the surface of a magnetic rotor. Among these methods, the improved adaptive median filtering method was used to filter out the image interference caused by the system. Image sharpening, feature point extraction, and expansion algorithms based on mathematical morphology were all used to identify the cracks. FCEA refined the feature point lattice into feature curves, and ROA enhanced the ability to detect fracture cracks. An SVM classifier trained by eight features of the feature curve was used to distinguish the crack curve from the interference curve.

The experiments illustrate that the algorithm proposed in this paper can effectively identify fracture cracks that cannot be identified by conventional algorithms. The experiments also show that the algorithm possesses a fast execution time. The accuracy of recognition was 97.9% using ROA, and the accuracy of recognition was 94.7% using IROA. However, the IROA algorithm execution time was 3% of ROA's execution time, and IROA's average crack detection time was shorter than that of ROA by 51%. In addition, we found that the method needed advanced optimization in certain parts of the algorithm. The first optimization procedure involved applying principal component analysis (PCA) to those features with strong correlation. This optimization procedure minimizes the redundancy and improved our algorithm's speed. Finally, this paper studied crack detection on simple shapes. To identify crack detection on complex morphologies (such as net cracks and circular cracks), further research is warranted to understand the law of crack extension paths and the features of different parameters of complex cracks.

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